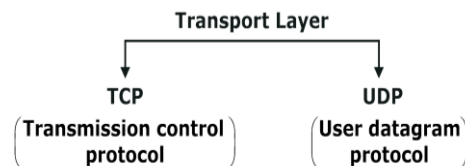


5

TCP & UDP

5.1 Introduction

Transport Layer can be connection oriented or connection less.



1. TCP is reliable process to process delivery of entire message.
2. TCP is a connection oriented.
3. TCP connection are full duplex and point to point.
4. TCP connection has 3 phases
 - i. Connection establishment
 - ii. Data transfer
 - iii. Connection termination
5. Each TCP connection is identified uniquely by Source Port + Destination Port + Source IP + Destination IP.
6. Each TCP connection is associated with Four window.
7. TCP uses three way “Handshake” to establish TCP connection.
8. TCP is not useful for Broadcasting and Multicasting.
9. TCP Header size is 20 bytes but if option (40 bytes) is added it will become 60 bytes.
10. TCP provide end to end error control and flow control.
11. Data will be received at the destination in order.
12. Data may arrive out of order and be temporarily stored by receiving TCP, but TCP guarantee that no out of order data delivered to the process.

5.2 TCP header

Source Port (16 bits)								Destination Port (16 bits)							
Sequence number (32 bits)															
Acknowledgement number (32 bits)															
Header Length (4 bits)	Reserved bits (6 bits)	U	A	P	R	S	F	Window Size (Advertisement Window) (16 bits)							
		R	C	S	S	Y	I								
		B	K	H	T	N	N								
Check sum (16 bits)								Urgent Pointer (16 bits)							
Options (0 - 40 bytes)															

Port Number	Name
0 – 1023	Well known port No.
1024 – 49151	Registered Port No.
49152 – 65535	Dynamic Port No.

$$\text{Wrap Around time (WAT)} = \frac{\text{Total sequence No.}}{\text{Bandwidth [Bytes / Sec]}}$$

- Minimum sequence number required to Avoid wrap Around time with in Life time = $B \times LT$
- Minimum number of bits required to Avoid wrap Around time with in LT = $\lceil \log_2 B * LT \rceil$

SYN = 1 → Consume one sequence number.

Ack = 1 → Consume zero sequence number.

FIN = 1 → Consume one sequence number.

1 Data byte → Consume one sequence number.

SYN	Ack	Meaning
1	0	request
1	1	reply
0	1	Ack
0	0	Data

Time out timer in TCP

Basic Algorithm	Jacobson's Algorithm
Time Out Timer = $2 * RTT$ Next Round Trip Time (NRTT) = $\alpha(IRT T) + (1 - \alpha)ART T$ $0 \leq \alpha \leq 1$	Time Out Timer = $4 * ID + RTT$ Next Round Trip Time (NRTT) = $\alpha(IRT T) + (1 - \alpha)ART T$ $0 \leq \alpha \leq 1$ Actual Deviation (AD) = $ IRT T - ART T $ Next Deviation (ND) = $\alpha(ID) + (1 - \alpha)AD$

5.3 Congestion Control

$$1. W_s = \min (W_C, W_R)$$

Slow start	Congestion Avoidance	Congestion Detection
1. If ACK Arrives $W_c = W_c + 1$	1. IF ACK Arrives $W_c = W_c + \frac{1}{W_c}$	1. Time out
2. After one RTT $W_c = 2 * W_C$	2. After one RTT $W_c = W_c + 1$	2. 3 duplicate ACK

5.4 Token Bucket

- Token bucket algorithm allows ideal hosts to accumulate credit for the future in the form of tokens.
- In regular interval, tokens are thrown into the bucket.
- Bucket has a maximum capacity.
- If there is a ready packet a token is removed from bucket and packet is sent.
- If there is no token in the bucket, the packet cannot be sent.

Let the capacity of token bucket is 'C' token and token enter into the bucket at rate of 'r' tokens per second. The maximum number of packet that can be enter into the network during the time interval 't' is

$$\begin{aligned}
 \text{Maximum number of packet} &= C + rt \\
 \text{Maximum average rate for token bucket } M &= \frac{C + rt}{t} \\
 Mt &= C + rt \\
 Mt - rt &= C \\
 (M - r)t &= C \\
 t &= \frac{C}{M - r}
 \end{aligned}$$

C → token Bucket capacity
r → Token Arrival rate

5.5 UDP

1. UDP is message oriented connection less Datagram protocol.
2. It is unreliable Transport protocol.
3. It does not provide Flow control and Error control & congestion control.
4. It does not add anything to the services except process to process delivery of data.
5. Header is simple and fixed in size i.e. 8 byte.

UDP Header

Source port (16 bit)	Destination port (16 bit)
Length (16 bit)	Checksum (16 bit)

Note:

Unlike TCP, the checksum calculation is not mandatory in UDP. No error control or flow control is provided by UDP. Hence UDP depends on IP and ICMP for error reporting.

5.5.1 Optional inclusion of checksum

The sender of UDP packet can choose not to calculate the checksum. In this case the checksum field is filled with all 0's before being sent.

5.5.2 Need of UDP

- [1] The application that required one request one reply. TCP is not suitable hence we use UDP.
- [2] Application that required constant dataflow TCP is not suitable hence we use UDP.
- [3] Application that required multimedia data transfer we cannot use TCP hence we use UDP.
- [4] Application that required fastness and then reliability TCP is not suitable hence we use UDP.
- [5] UDP used for management process such as SNMP (simple network management protocol).
- [6] UDP is used for some route updating protocol such as RIP.
- [7] For broadcasting & multicasting application TCP is not suitable hence we use UDP.
- [8] UDP is normally used for interactive real time applications.
- [9] UDP is suitable for a process with internal flow and error control mechanisms. For example, the Trivial File Transfer protocol(TFTP) process include flow and error control. It can easily use UDP

TCP	UDP
Connection-oriented	Connectionless
Reliability in delivery of message	Not reliable
Sequence Number.	No sequence number.
ACK number.	No ACK number.
Overhead is high(Header size 20-60 Bytes)	overhead is less(Header size = 8 Bytes)
Keep track of order (sequence)	No order
Protocols: HTTP, FTP, SMTP, POP	Protocol: DNS, SNMP, TFTP, DHCP, All real time protocol

